

SNI

The idea behind this section of the lab is to demonstrate filtering via keyword or Server Name Indicator (SNI). The setup once again involves using Windows 10 as our censored VM and Kali (set up to forward traffic) as our censoring VM. Using Telegram as our circumvention tool, we can start with the command on Kali

```
tshark -i eth1 -Y "tls.handshake.extensions_server_name contains \"t.me\"" -T fields -e ip.dst -e tls.handshake.extensions_server_name
```

This monitors the eth1 interface for the SNI "t.me" and extracts the destination IP and the server name. With it we detect t.me in the encrypted part of the TLS handshake.

We can use the same command for "telegram.org"

```
tshark -i eth1 -Y "tls.handshake.extensions_server_name contains \"telegram.org\"" -T fields -e ip.dst -e tls.handshake.extensions_server_name
```

Both VMs are started and then t.me and telegram.org are accessed via the browser. Interestingly enough, the t.me script generated packets when run, these could then be analyzed. The telegram.org script showed 0 packets, though.

Attempting to block t.me and telegram.org using iptables also showed an interesting result. Here are the iptables commands used:

```
sudo iptables -A OUTPUT -p tcp --dport 443 -m string --string "t.me" --algo bm -j DROP
sudo iptables -I OUTPUT -p tcp --dport 443 -m string --string "telegram.org" --algo bm -j DROP
sudo iptables -I FORWARD -p tcp --dport 443 -m string --string "t.me" --algo bm -j DROP
sudo iptables -I FORWARD -p tcp --dport 443 -m string --string "telegram.org" --algo bm -j DROP
```

When we go back to Windows, access to t.me is blocked, but telegram.org is reached easily. Why?

The answer lies in the use of QUIC. The handshake used by telegram.org used QUIC while t.me used the standard unencrypted TLS Handshake. QUIC used UDP, and iptables fails to block it. The only way around this is to block QUIC UDP port 443

```
sudo iptables -A OUTPUT -p udp --dport 443 -j DROP
```

This forces the connection to fall back to https, and it is subsequently blocked. An additional issue with trying to block using just an iptables match is that the string must match perfectly, or the connection isn't

blocked. It is not a robust censorship method.